



Event Triggers

<Practice with exploring event triggers concept>



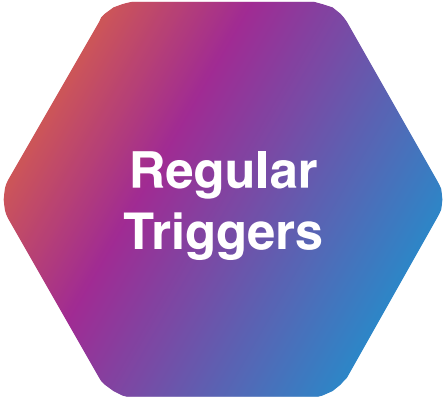
About me:

- Java Developer
- Co-Organizer of Codeus community

What is event trigger?

Event trigger - a special type of trigger that fires on database-level events (DDL operations), not on table data changes.

Office Security Analogy



**Regular
Triggers**

Security cameras in rooms (Monitor what happened in specific rooms)



**Event
Triggers**

Building access control system (Monitor who enters/exits, opens/closes doors)

Regular triggers

vs

Event triggers

Respond to data manipulation

Response

Respond to DB structure changes

Table-specific and localized

Scope

Have database-wide visibility and control

Can fire before, after, or instead of operations

Timing

Fire before or after entire DDL statements

Handle business logic and data validation

Use cases

Auditing, security, and schema governance

Event Trigger Types

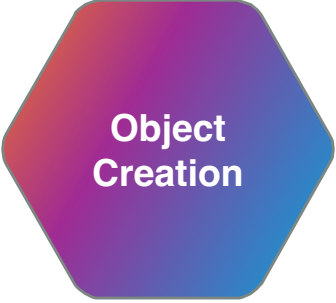
ddl_command_start - BEFORE DDL execution

ddl_command_end - AFTER DDL execution

table_rewrite - When table is rewritten (PostgreSQL)

sql_drop - When objects are dropped

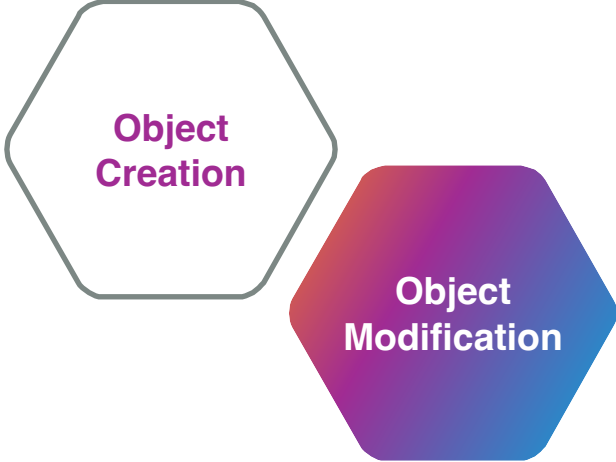
DDL commands that fire event triggers



Object
Creation

CREATE TABLE, CREATE INDEX, CREATE VIEW, CREATE FUNCTION

DDL commands that fire event triggers

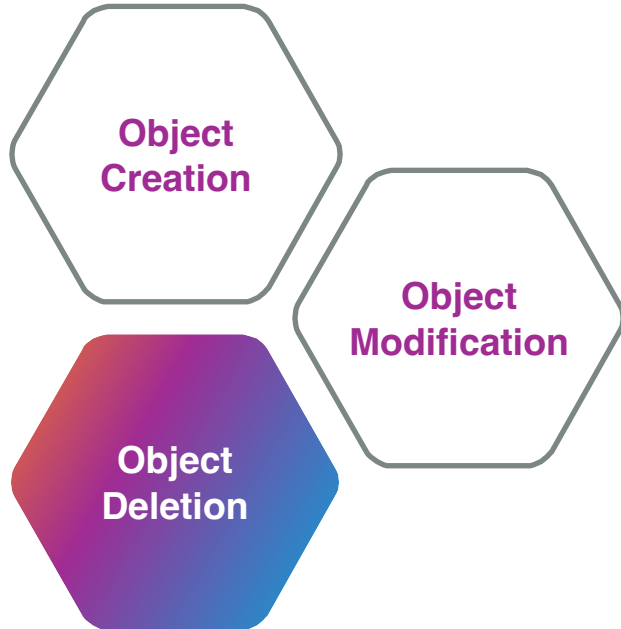


Object
Creation

Object
Modification

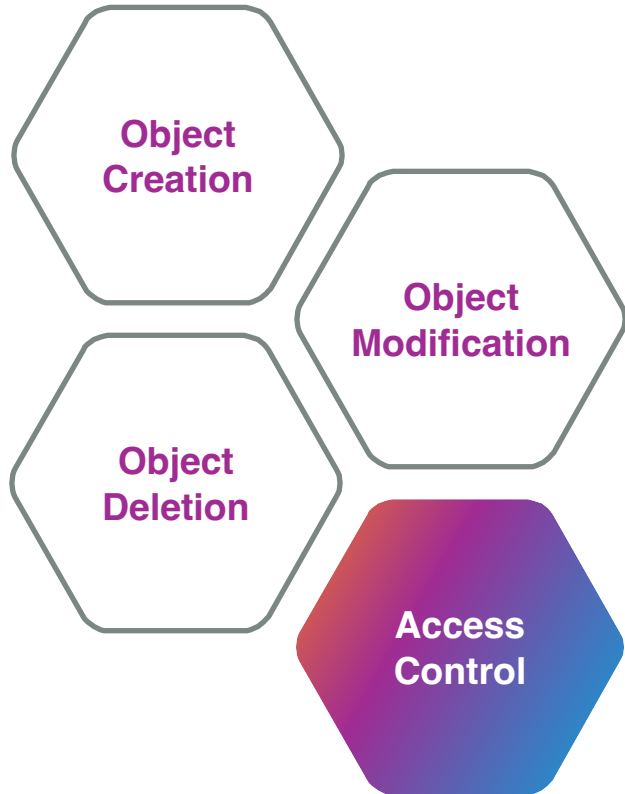
ALTER TABLE, ALTER INDEX, ALTER VIEW, ALTER FUNCTION

DDL commands that fire event triggers



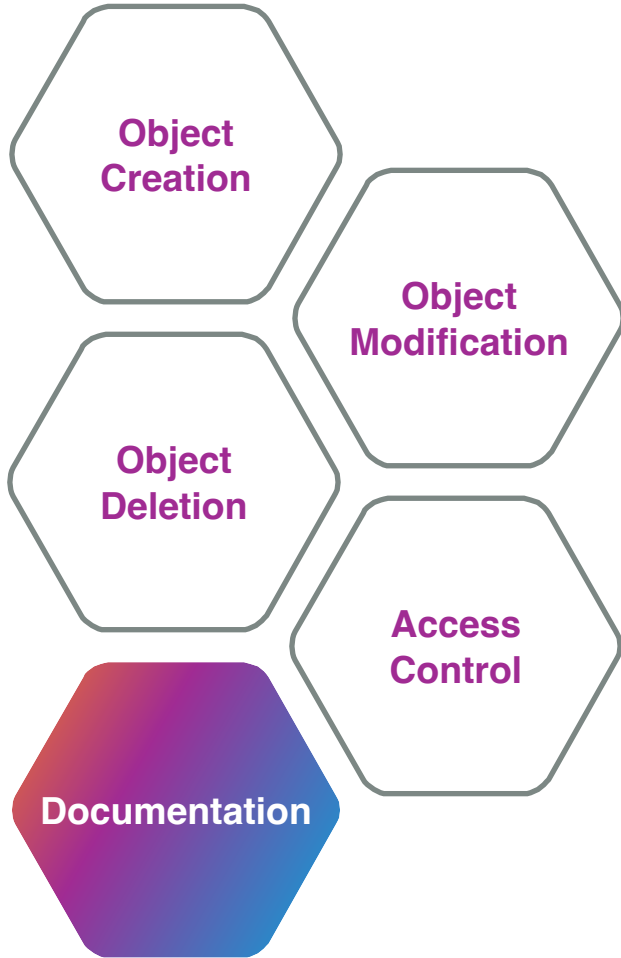
DROP TABLE, DROP INDEX, DROP VIEW, DROP FUNCTION

DDL commands that fire event triggers



GRANT, REVOKE

DDL commands that fire event triggers



COMMENT ON TABLE, COMMENT ON COLUMN

When to use event triggers



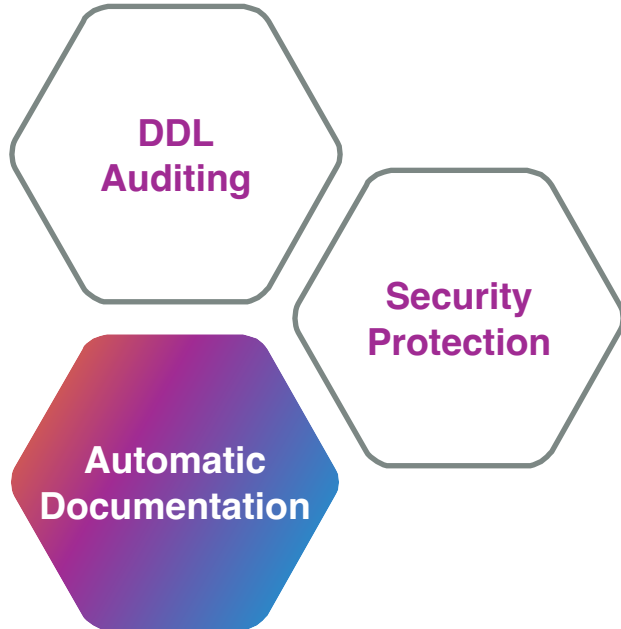
"Who created/modified/deleted what and when?"

When to use event triggers



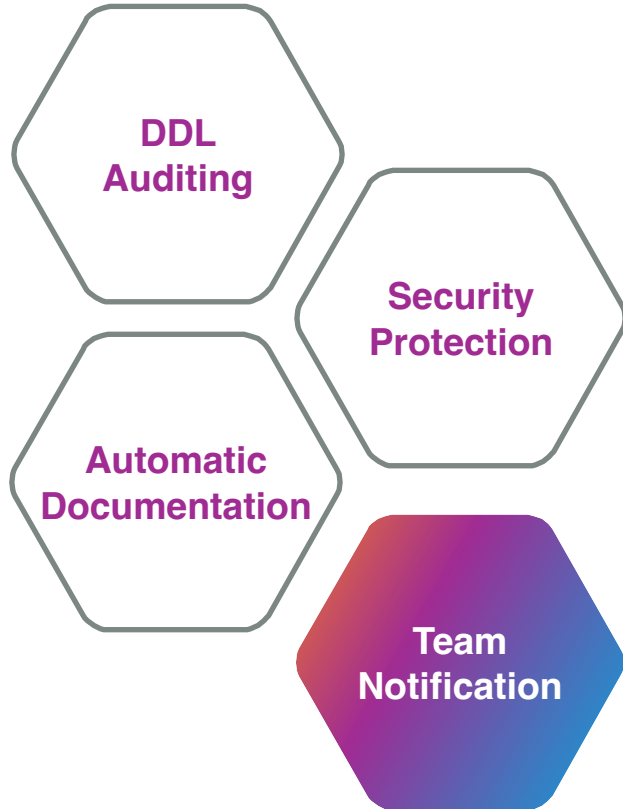
"Prevent accidental DROP operations in production"

When to use event triggers



"Auto-comment new tables with creation info"

When to use event triggers



"Notify Slack when schema changes happen"

Basic Event Trigger Example

Step 1: Create the Function

sql

```
CREATE OR REPLACE FUNCTION simple_ddl_log()  
    RETURNS event_trigger  
    LANGUAGE plpgsql  
AS $$  
BEGIN  
    RAISE NOTICE 'DDL executed: % by user %', tg_tag, session_user;  
END;  
$$;
```


Basic Event Trigger Example

Step 2: Create the Event Trigger

sql

```
CREATE EVENT TRIGGER ddl_notifier  
ON ddl_command_start  
EXECUTE FUNCTION simple_ddl_log();
```

Basic Event Trigger Example

Step 3: Test It

sql

```
CREATE TABLE test_table (id SERIAL);
```

```
-- Result: NOTICE: DDL executed: CREATE TABLE by user postgres
```

```
DROP TABLE test_table;
```

```
-- Result: NOTICE: DDL executed: DROP TABLE by user postgres
```

Event Trigger Limitations

**What Event Triggers
CAN DO**

Read data from tables

Log events and metadata

Block operations (via `RAISE EXCEPTION`)

Get comprehensive DDL operation details

Event Trigger Limitations

What Event Triggers CANNOT DO

Modify table data (read-only access)

Use transaction commands (`COMMIT`, `ROLLBACK`)

Work with temporary tables

Fire on system catalog operations

Best Practices Using Event Triggers

DO

Keep functions simple - Event triggers should be fast

Log rather than block - Better to record than prevent

Handle exceptions - Always use `EXCEPTION` blocks

Test thoroughly - Verify on test databases first

Document purpose - Comment why each trigger exists

Best Practices Using Event Triggers

DON'T

Complex logic - Keep trigger functions simple

Long operations - Avoid time-consuming calculations

Circular dependencies - Triggers calling themselves

Data modifications - Event triggers are for monitoring

Thank you

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